Multi-view Semantic Assessment

Scene description & views

"A cozy reading den with a tall wooden bookshelf against one wall, a leather armchair beside it, a small side table holding a hardcover book, an arc floor lamp illuminating the seat, and a potted fern near the window."



Top-down

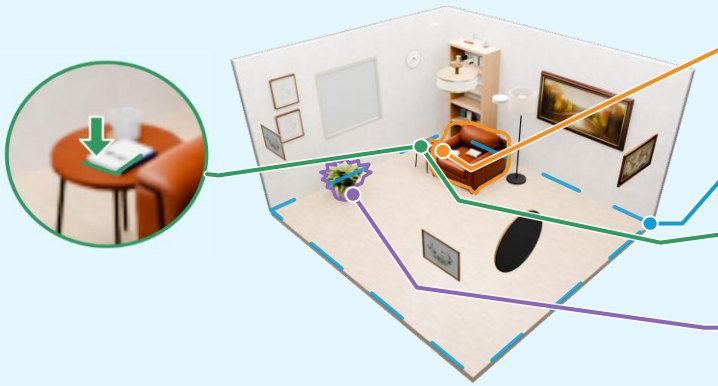


Corner views

MLLM semantic assessment

Realism	How realistic the room appears?
Functionality	How well the room supports the intended activities?
Layout	How well is the furniture arranged and aligned with user preferences?
Completion	How complete and finished the room feels?

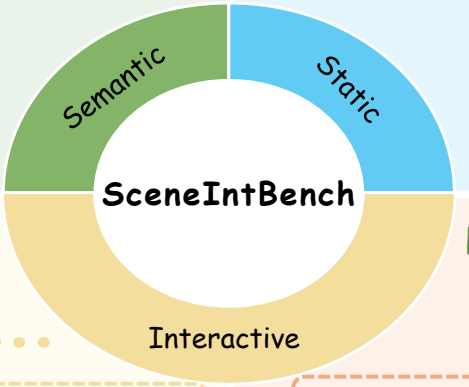
Static Physical Plausibility



Static checks

- **Collision** — Inter-object penetration
- **Out-of-Bounds** — Room-boundary containment
- **Support** — Contact and support
- **Stability** — Physically stable at rest

Check geometry and physical validity at rest



Active Interaction Evaluation

GPT-5.6 Sol: <Infer applicable actions from the scene description.>

Action set: Push chair | **Push plant** | Place item on table | ...

Action execution

Isaac Sim

Instruction: Push the potted plant 0.15m and keep it upright for 1s.



Initial state



Interaction



Post-action

Interaction metrics



Simulator states

Action video

Rule-based	ISR	Task success
	PIV	Post-action violation
MLLM judge	ICP	Capability & progress
MLLM judge	DP	Dynamic plausibility

Plausible at Rest

Physically Unreliable in Action

Missing articulation

Open fridge

Visible door ≠ operable door



Contact inconsistency

Push plant

Passed static checks, failed during contact



Post-action violation

Push chair

Non-target support loss and displacement

